Tutorial-8

Question 1

Given

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix},\ P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Find PA. Also state the order in which the rows of Ahave been permuted**.**

Question 2

Using the Gauss-Jordan elimination method, find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 3 & 5 & 4 \end{bmatrix}.$$

Question 3

Determine whether the following matrix is invertible:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 3 & 4 \end{bmatrix}.$$

If the inverse does not exist, give a suitable reason.

Question 4

Given

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix},$$

find $A^{-1}$and A^{T}.

Hence, verify that

$$\boxed{(A^{-1})^{T} = (A^{T})^{-1}}.$$

Question 5

Consider a point P = (2, 3).

(a) Find the transformation matrix for a rotation of 90 degrees anticlockwise about the origin.

(b) Find the transformation matrix for a horizontal shear with shear factor 2.

(c) Find the transformation matrix for scaling the x-coordinate by 3 and the y-coordinate by 2.

Question 6

Consider the matrix

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix}.$$

Determine whether A^{-1}exists.

Justify your answer using following methods:

1. Determinant
2. Rank
3. Linear dependence of rows/columns
4. Gauss-Jordan elimination
5. Singular matrix definition

QUESTION 7

Let A be the matrix:

A = [1 2]
[2 5]

(a) Using the Gauss-Jordan elimination method, find the inverse of matrix A.

(b) Find the transformation matrix for a horizontal shear with shear factor 3.